Pedestal Family: Loss & Overlap Breakdown

`shg_mode_survey · pedestal1 + pedestal_sweep1 + pedestal_vertical1`

416 refined pump→SH crossings across 51 geometries. Sub-metrics that eta_pct_per_W_cm2 collapses into one number: scattering vs. absorption at both wavelengths, raw mode-shape overlap independent of loss, and how little either one predicts the other.

SH = 215–235 nm, deep UV. Pump = 430–470 nm, true blue. Two very different bands, two very different loss stories.

CAUTION

This is the 'pedestal' family (AlN ridge stacked on TOP of an SiNx rib) -- the structure every 'hybrid'-named run before 2026-10-10 actually built by accident, due to a geometry-construction bug (see PLAN.md). It is a real, different waveguide, but it is NOT the Honda et al. AlN-embedded-in-SiNx structure the 'hybrid' family now (after the fix) correctly builds. AlN is modeled as lossless here (its own UV k is a separate, still-open item) -- every absorption number below is SiNx bulk absorption.

SH ABSORPTION / SCATTERING
768× median
SiNx bulk absorption dwarfs sidewall scattering in the UV

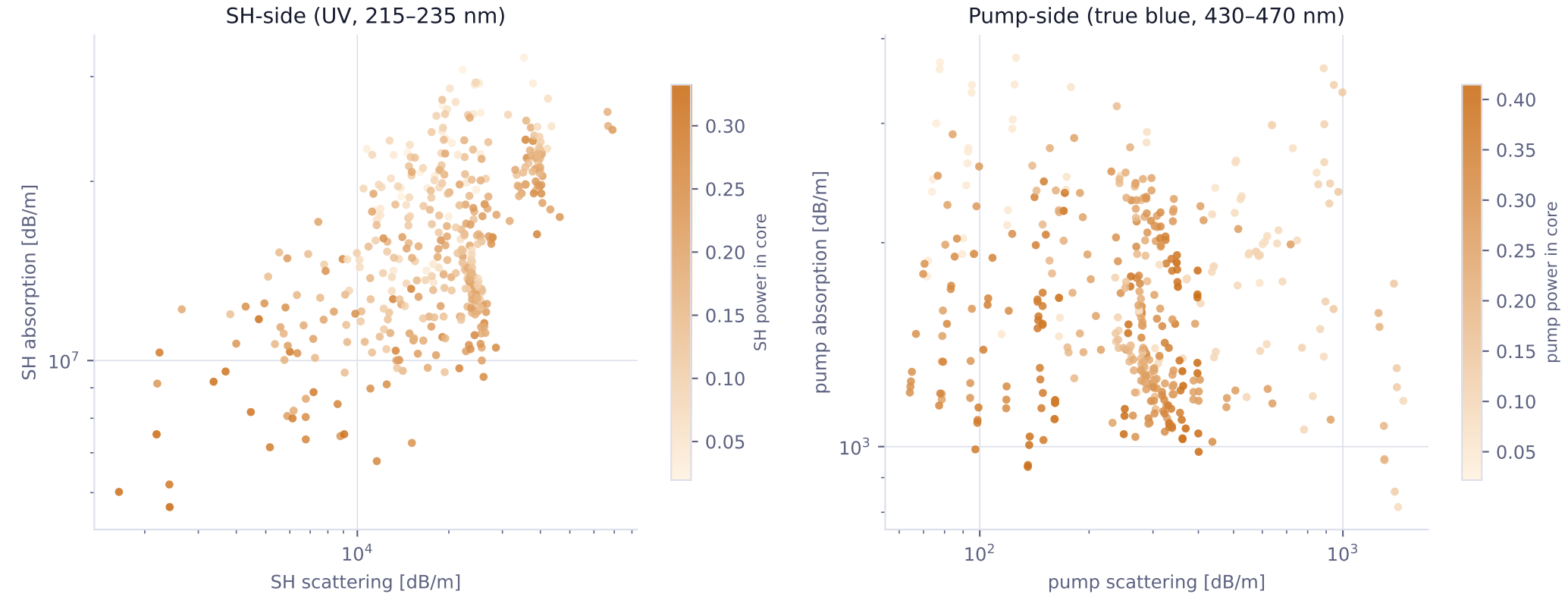
SH LOSS / PUMP LOSS
8104× median
same crossing, UV vs. blue -- very different bands

OVERLAP_SHAPE ↔ CONFINEMENT
r = -0.62
narrow cores: highest shape overlap, least SH power actually in them

OVERLAP_SHAPE ↔ LOG(ETA)
r = 0.02
shape overlap alone barely predicts final efficiency

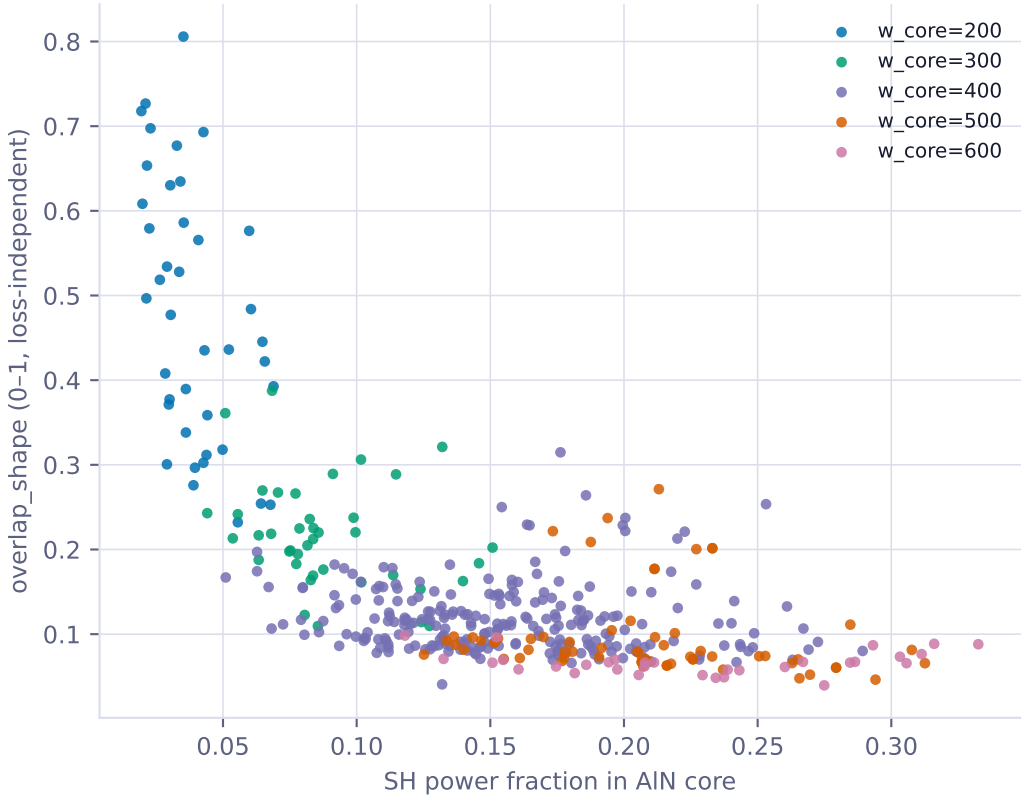
`Best crossing overall: TM01→TM81 @ 233.60 nm, eta=35.2 %/W/cm² (h=300/w=400/sin_w=700/sin_h=500, run=pedestal_vertical1)`

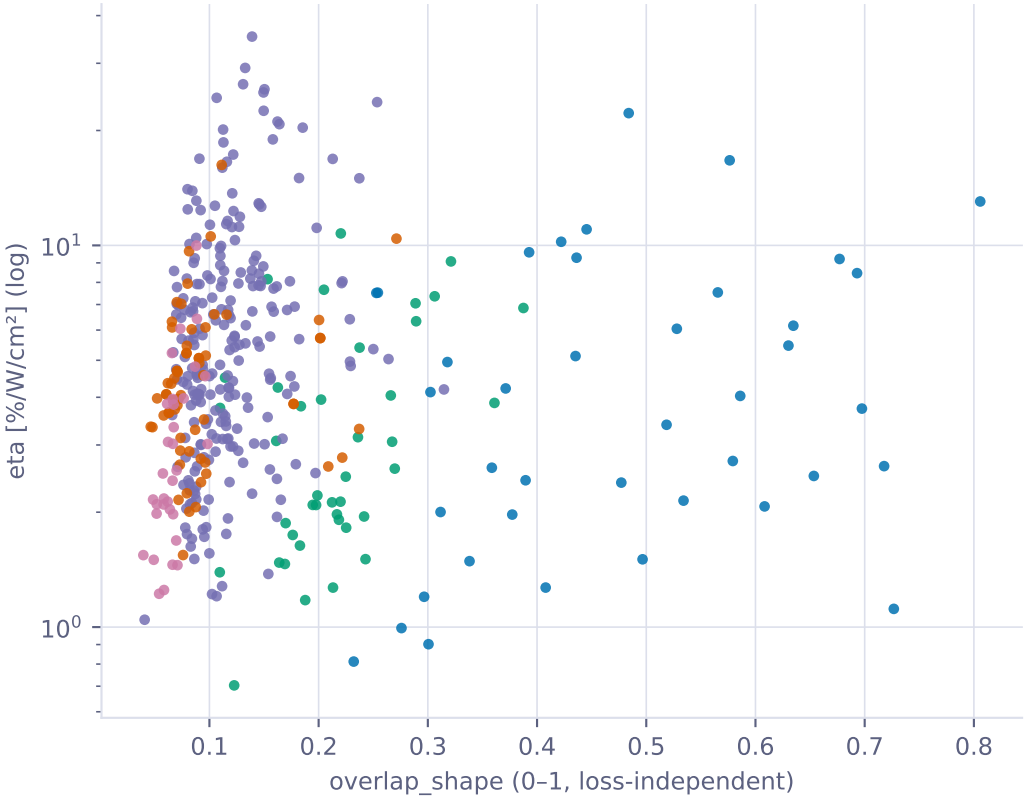
Loss breakdown: scattering vs. absorption



Note the axis scales: SH absorption sits 3 orders of magnitude above SH scattering; pump loss overall is ~1000x smaller than SH loss.

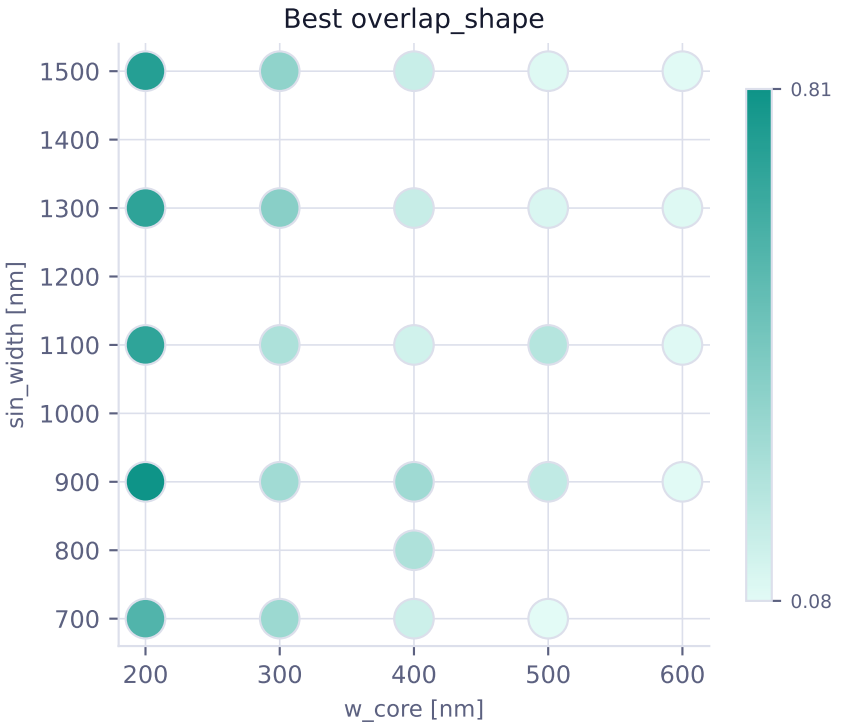
Overlap vs. confinement vs. final efficiency

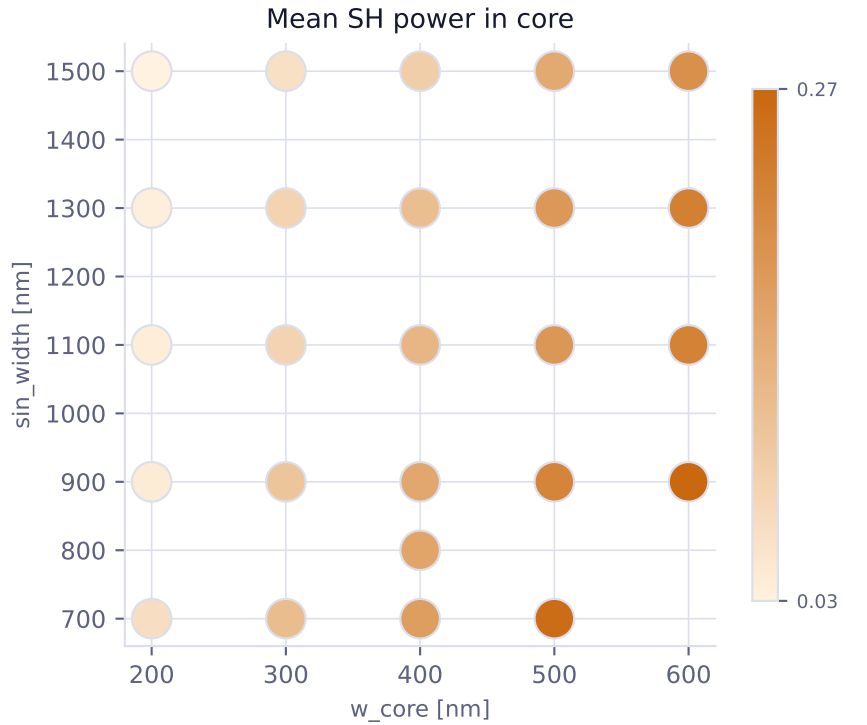


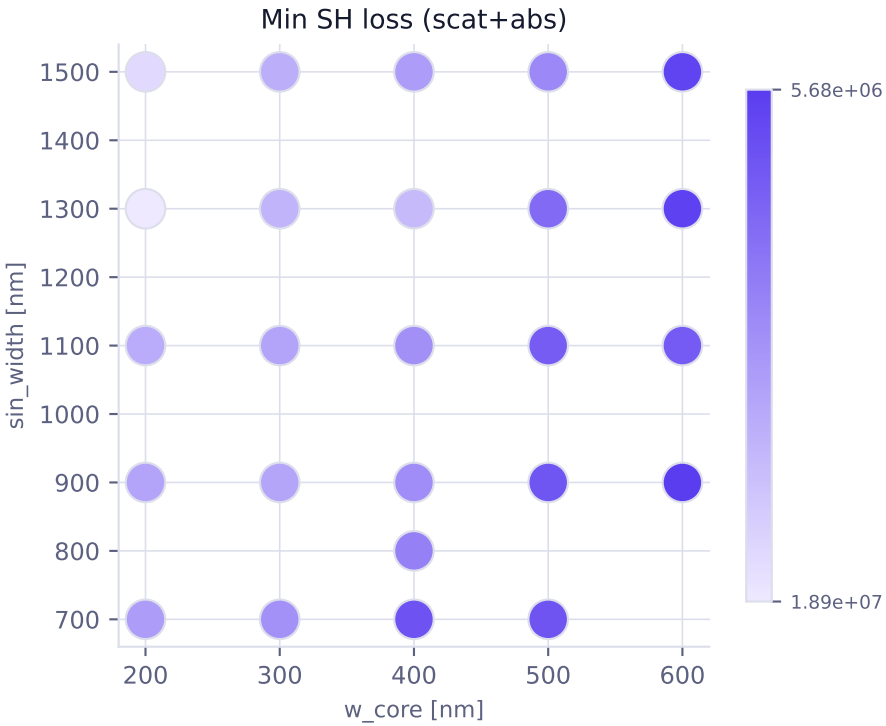


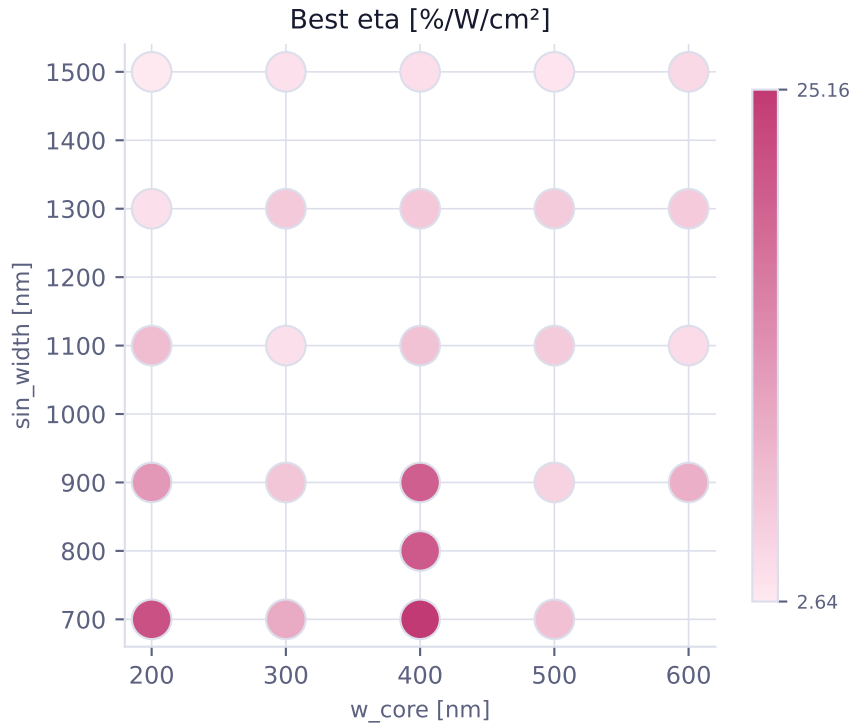
Lateral geometry maps

Fixed h_core = 400 nm



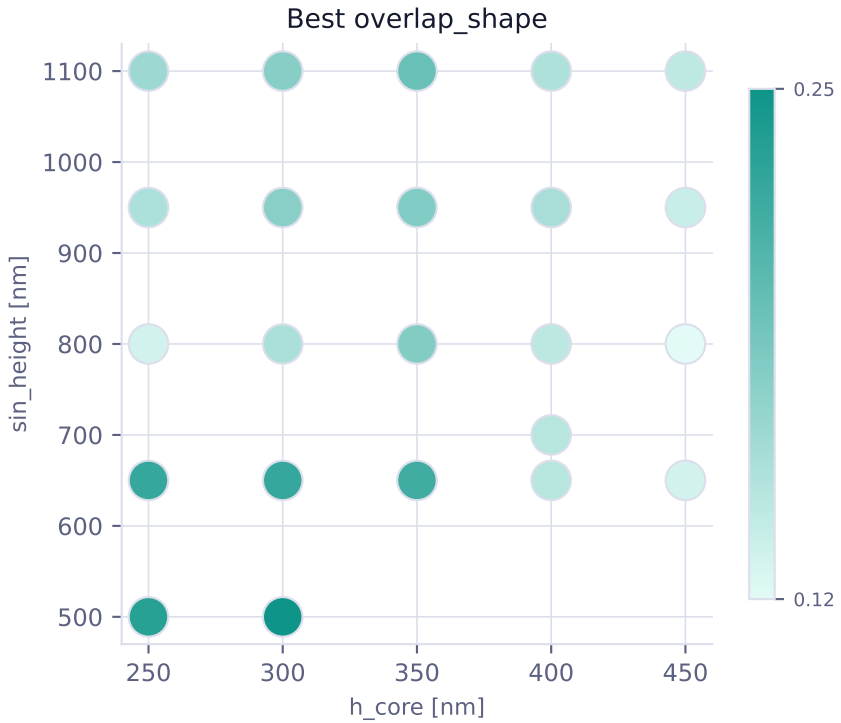


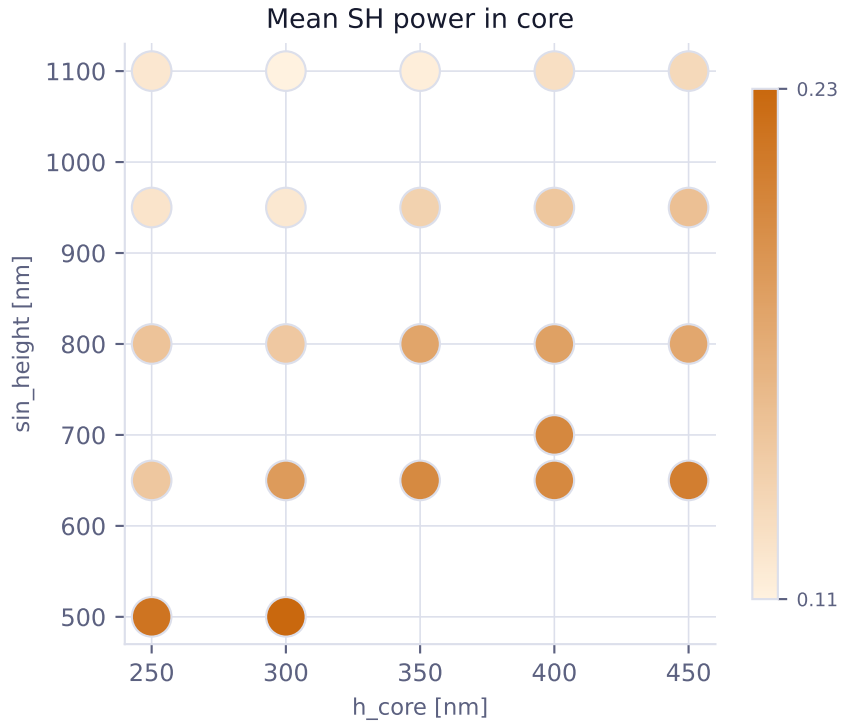


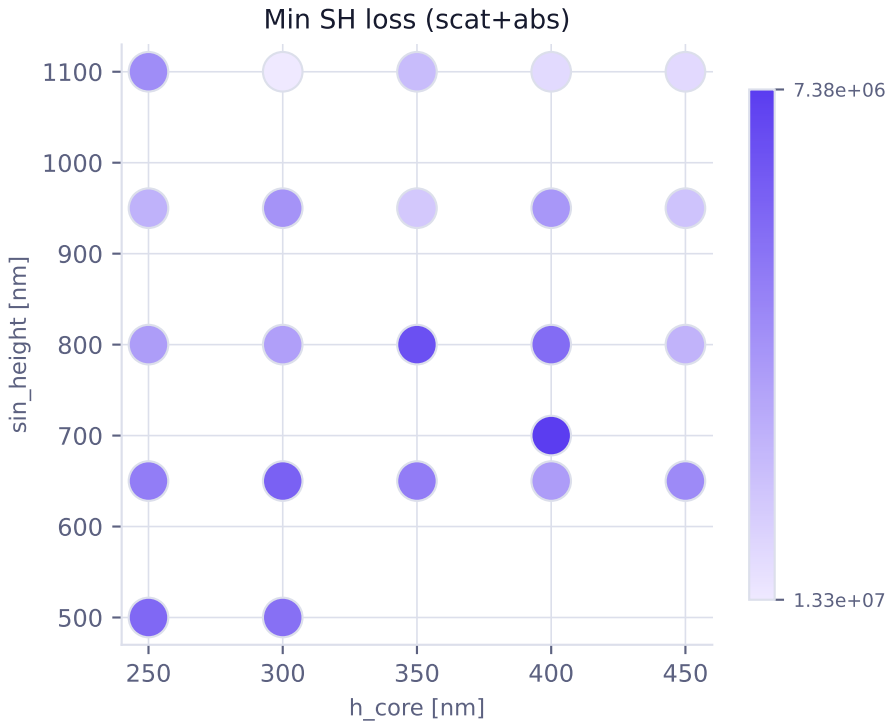


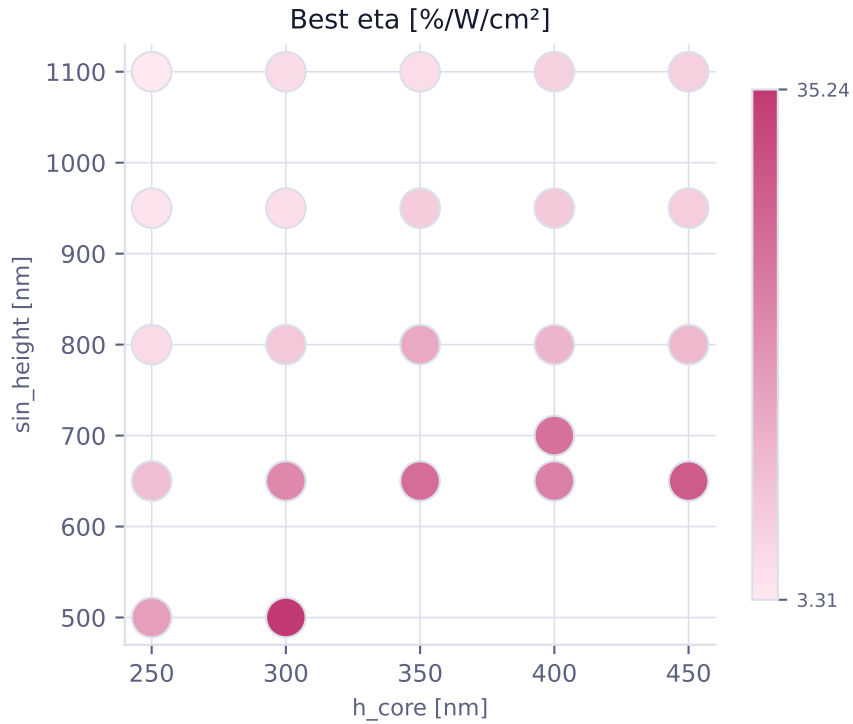
Vertical geometry maps

Fixed w_core = 400 nm, sin_width = 700 nm

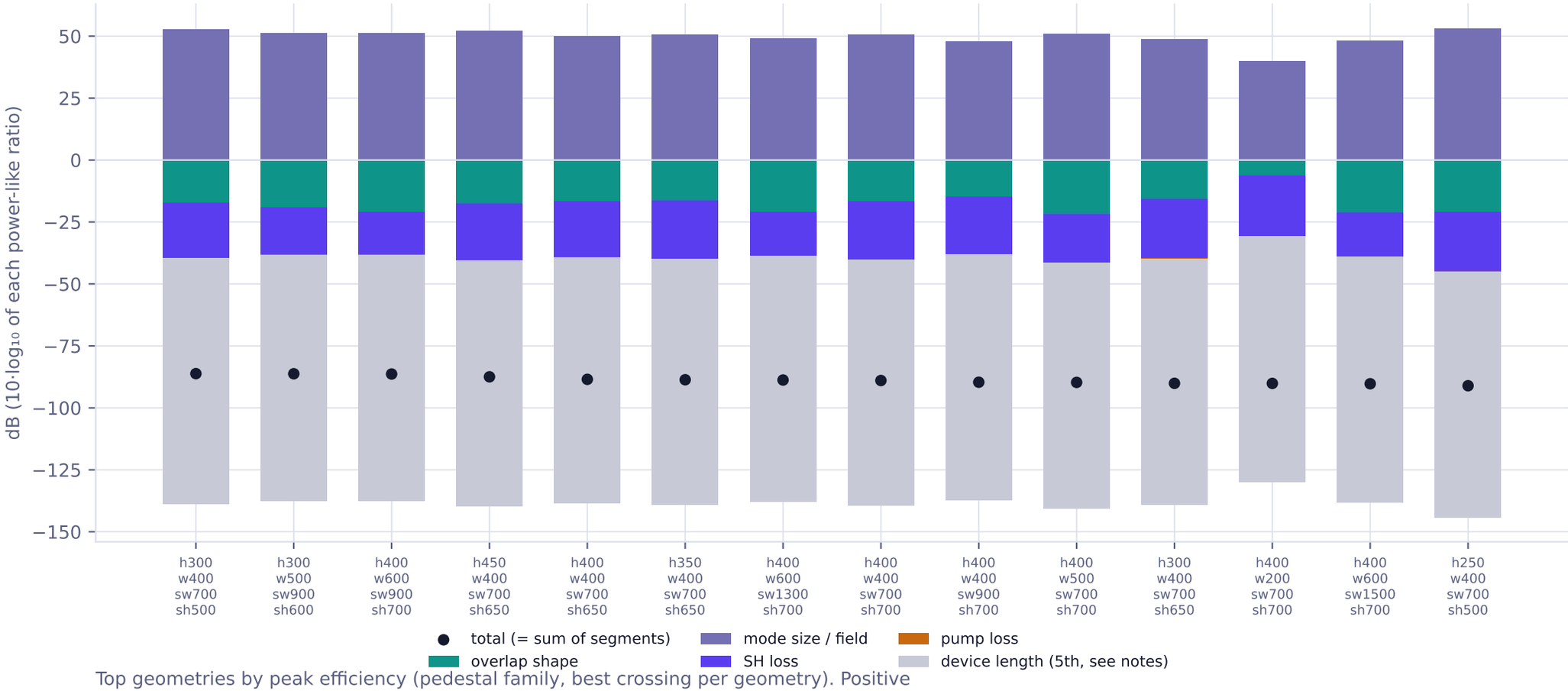








NCE breakdown: dB-additive waterfall per geometry



Top geometries by peak efficiency (pedestal family, best crossing per geometry). Positive segments stack upward, negative downward, from 0 -- the black dot always lands exactly on their sum. 'Mode size / field' is the one unbounded segment (a physical scale set by confinement and the d-tensor, not a 0-1 quality factor like the other three); everything else is <= 0 dB by construction (overlap_shape <= 1, loss always reduces L_eff).

'Device length' is a 5th, unrequested-but-necessary term: each bar is evaluated at ITS OWN optimal propagation length (loss_limited_length's L_opt, same quantity behind this project's usual peak_efficiency_pct_per_W) rather than one shared length -- without it the other four terms alone would not sum to the true total. See nce_decomposition.py's docstring for the two earlier (rejected) versions that didn't reconstruct the total on real data.

Notes & provenance

Data: runs/pedestal1, runs/pedestal_sweep1, runs/pedestal_vertical1 crossings.csv, refined rows only (screening doesn't solve losses or overlap_shape -- too expensive per-crossing to run on every screened candidate). Extracted via extract_subfigure_data.py, rendered via build_subfigure_report.py.

overlap_shape is dimensionless 0-1, integrated over the AlN region only -- 1 means the SH field inside AlN is exactly proportional to the pump-driven polarization; it carries no information about HOW MUCH SH power actually reaches AlN (that's sh_power_in_core, measured separately).

Absorption numbers are SiNx shell bulk absorption (Gamma_shell * alpha_SiNx(lambda), first-order perturbation -- survey.py's bulk_absorption_loss()); AlN is modeled as lossless throughout, so it contributes exactly zero. SiNx's own measured k rounds to 0 in the raw literature data at 430-470 nm (pump/blue) but is 0.06-0.14 at 215-235 nm (SH/UV) -- see materials/fit_sinx_sellmeier.py.

All 416 refined crossings here are clean by the window/box-mode check (sh_edge_ratio well below the 0.02 threshold throughout).

IMPORTANT: this is the 'pedestal' family -- AlN stacked on top of an SiNx rib, not embedded in it. It does not test the Honda et al. 2025 TM00->TM20 mechanism. See PLAN.md's 2026-10-10 and 2026-10-12 entries for the full story, and the corrected 'hybrid' family for the structure that does test that mechanism (re-run pending as of this report).